

Multiple Variable Calculus

6. Partial Derivatives

1. Given $u = x^2 + 2x - 1$; $x = t^2 - 1$

Find $\frac{\partial u}{\partial t}$

Solution: $\frac{\partial u}{\partial x} = 2x + 2$

$$\frac{\partial x}{\partial t} = 2t, \quad = 2(t^2 - 1) + 2 = 2t^2$$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial t} = (2x + 2) \cdot 2t = 4xt + 4t$$

2. $u(x, y) = \tan x / y$

$$f(r, \theta) = r^2 \sin^2 \theta + r^3$$

$$u(r, s, t) = r^3 + s^2 t + (t-1)(r-3)$$

$$f(p, \varepsilon) = e^{p^2 \log \varepsilon}$$

$$(i) \frac{\partial u}{\partial x} = \sec^2 \frac{x}{y} \cdot \frac{1}{y} = \frac{1}{y} \sec^2 x/y$$

$$\frac{\partial u}{\partial y} = \sec^2 \frac{x}{y} \left(\frac{-x}{y^2} \right) = \frac{-x}{y^2} \sec^2 x/y$$